Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 3: Lumped Elements, Characteristics and the LTI Resistor

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Colour code used in these notes and in the slides			
charge q	current i	voltage v	power p, energy w

1. The other half of the equations

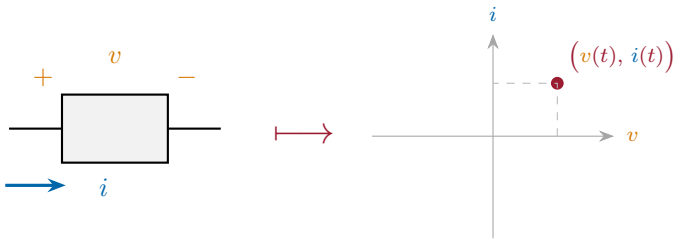
Lecture 2 gave us the *network* equations: KCL and KVL, which know how the elements are wired and nothing about what they are. The other half of the $2b$ equations is one **terminal equation** per element, and that is where all the physics hides.

Network equations	+	**Terminal equations**
KCL, KVL		one per element
done in Lecture 2		this lecture

So far only one element has been given a terminal equation — the linear resistor, $v = Ri$. This lecture asks the general question: *what can a two-terminal element be?*

2. An element is a curve in the v–i plane

Take any two-terminal element, attach the reference directions of the passive sign convention, and watch it. At each instant it has a voltage v and a current i — one point in a plane.

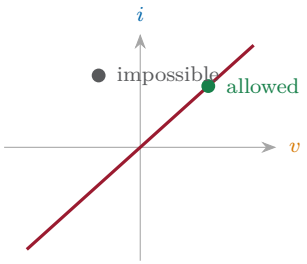


As time passes the point moves. The element is defined by *which points it is allowed to visit.*

Characteristic

The **characteristic** of a two-terminal element is the collection of *all* the pairs (v, i) that the element allows. Drawn in the plane, it is a **curve**: the operating point may sit anywhere on that curve and nowhere else.

Everything the element does is in that picture; nothing else about it matters to circuit theory. If the curve itself changes with time we write it as the curve *at time* t.



The element is a *rule* about which pairs may occur. Feed it a voltage and the curve tells you the current; force a current through it and the curve tells you the voltage.

Every element is a curve

A linear resistor is a straight line through the origin. A battery is a vertical line. A diode is a bent curve.

For every new element, we ask the same thing: *which curve?* Section 7 is about the shape of that curve and nothing else.

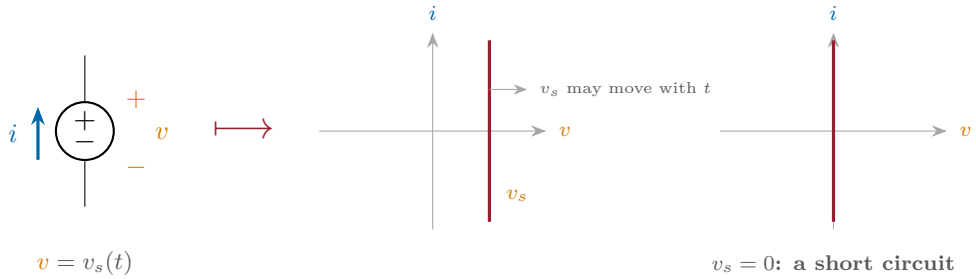
3. The independent voltage source

Independent voltage source

An element whose terminal voltage is a prescribed function of time, *whatever current flows through it*:

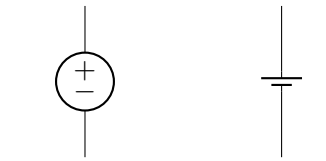
$$v(t) = v_s(t) \qquad \text{for every } i.$$

Its characteristic at time t is the **vertical line** $v = v_s(t)$.



$v = v_s(t)$ $v_s = 0$: **a short circuit**

3.1. Two symbols, one element



general source battery

Both are independent voltage sources and both appear in textbooks, in the laboratory and in this course. The battery symbol carries an extra hint — it is a *constant* source — while the circle covers any $v_s(t)$ at all.

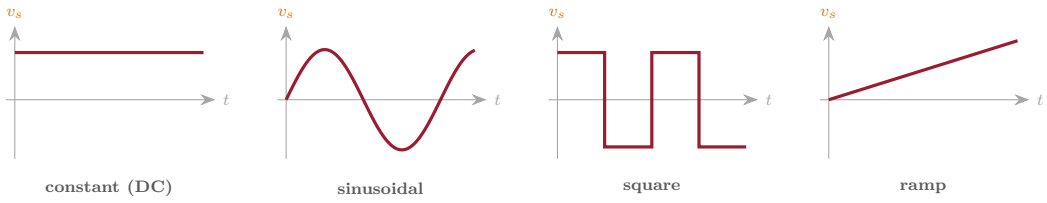
These notes use the **circle** from here on.

A voltage source does not decide the current

The line is vertical precisely because i is *unconstrained* by the element: the source will supply whatever current the rest of the circuit demands at that voltage. Which current actually flows is settled by KCL, KVL and the other elements — never by the source alone.

3.2. Waveforms

$v_s(t)$ is any function you like; these four cover most of this course.



Example 3.1 — the source does not act alone

A 12 V source is connected first to a 4 Ω resistor, then to a 2 Ω one.

$$i = \frac{12}{4} = 3\,\text{A}, \qquad i = \frac{12}{2} = 6\,\text{A}.$$

The source's characteristic never changed — it is the same vertical line at $v = 12\,\text{V}$. What changed is the *other* element's line, and with it the point where the two characteristics meet. Section 11 makes that picture explicit.

4. The independent current source

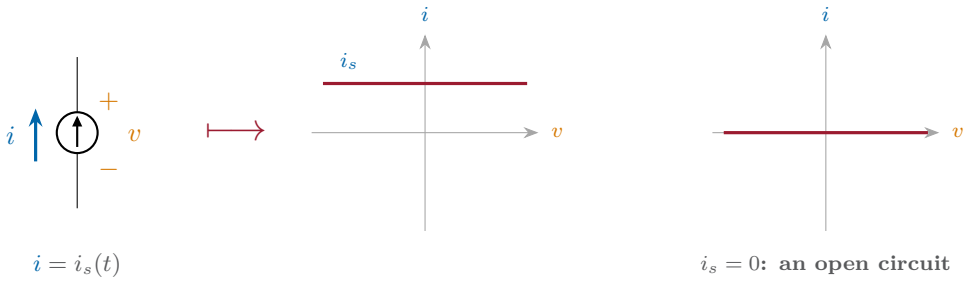
Everything above, with v and i exchanged.

Independent current source

An element whose terminal current is a prescribed function of time, *whatever voltage appears across it*:

$$i(t) = i_s(t) \qquad \text{for every } v.$$

Its characteristic at time t is the **horizontal line** $i = i_s(t)$.

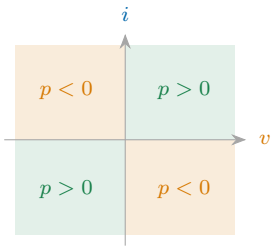


The dual warning

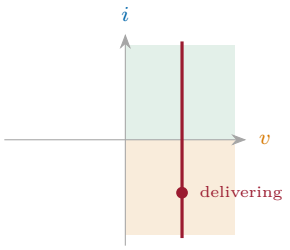
A current source does not decide its voltage. Connect 2 A to 4 Ω and it produces 8 V; connect it to 100 Ω and it produces 200 V; leave it open-circuited and the ideal model demands infinite voltage — which is the model telling you that a real current source has a limit, and beyond it something gives way.

5. Sources are active

With the passive sign convention, $p = vi$ is the power *absorbed*. So the sign of p is decided by which quadrant of the v–i plane the operating point sits in.



absorbing vs delivering



a 12 V battery driving a lamp

A characteristic that stays inside the green quadrants can only ever *absorb*. One that reaches into an orange quadrant can *deliver*, and the element is called **active**.

Passive and active

An element is **passive** if its characteristic lies entirely in the first and third quadrants at every instant, so that $p = vi \geq 0$ always. Otherwise it is **active**.

A voltage source with $v_s \neq 0$ is a vertical line that crosses both the first and the fourth quadrant, so it is active — as it must be, since driving a circuit means giving out energy. The same argument makes any current source with $i_s \neq 0$ active. Short circuits, open circuits and positive resistances are passive.

6. The resistor — a much larger family than you expect

Resistor

A two-terminal element is a **resistor** if, at every instant, its characteristic is a curve in the v–i plane.

Read that carefully: it says nothing about straight lines, nothing about Ohm's law, and nothing about R. What it does say is that the relation between v and i is *purely algebraic* — no derivatives, no integrals, no memory of what happened a moment ago. The present current depends only on the present voltage and vice versa.

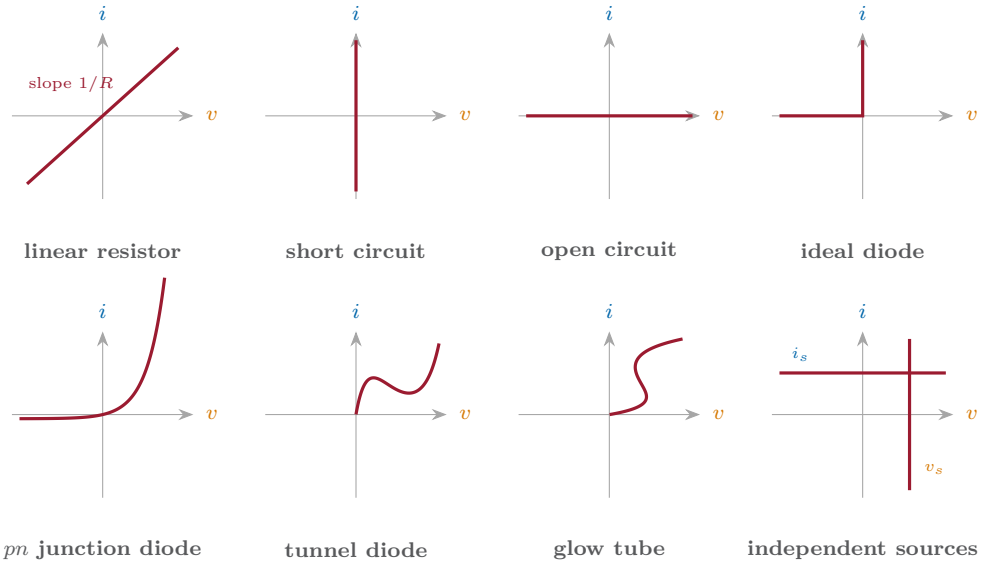
Consequences worth pausing over

- A *pn* junction diode is a resistor. So is a tunnel diode, a light bulb, a thermistor.
- A short circuit and an open circuit are resistors.
- An *independent source* is a resistor too — its characteristic is a straight line, just not through the origin.
- Capacitors and inductors are **not** resistors: $i = C\,dv/dt$ is not a curve in the v–i plane, because the same v allows any i depending on how v is changing.

The linear resistor of Lecture 1 is one member of this family — the straight line through the origin — and it is only the simplest.

Yes, that really does make a battery a resistor. The word here describes the *shape of the relation* — algebraic, memoryless — and says nothing about dissipating energy; active elements qualify just as well as passive ones. If it helps, read "resistor" as "memoryless element" every time.

6.1. A gallery



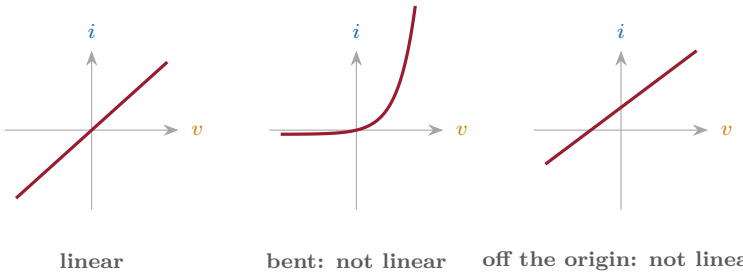
7. Five words for describing a resistor

Every one of them is a statement about the *shape* of the characteristic.

7.1. Linear

Linear

The characteristic is a **straight line through the origin**. Equivalently, superposition holds: scaling v by α scales i by the same α.

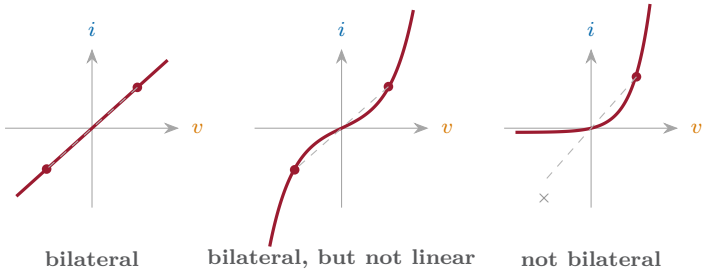


A straight line that does *not* pass through the origin is not a linear element, however straight it looks — a battery is the obvious example.

7.2. Bilateral

Bilateral

The characteristic is **symmetric about the origin**: whenever the pair (v, i) is on the curve, so is $(-v, -i)$. Physically: swapping the two terminals round makes no difference.



Every linear element is bilateral — but not conversely. Two diodes back to back give the middle curve: bent, yet perfectly symmetric.

A single diode conducts one way and blocks the other, and that is what *not bilateral* means.

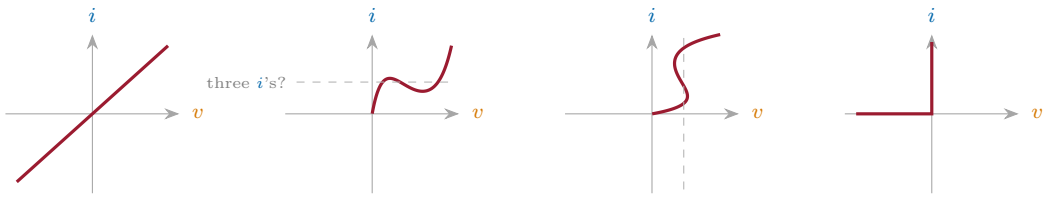
7.3. Active or passive

Defined in Section 5: passive when the characteristic never leaves the first and third quadrants, so the element can only absorb. Independent sources are active; a positive linear resistance, a short and an open are passive. A *negative* resistance — a straight line through the origin with negative slope — is a line in the second and fourth quadrants, and so is active.

7.4. Voltage-controlled and current-controlled

Controlled by which variable?

A resistor is **voltage-controlled** if its characteristic can be written $i = f(v)$ — one current for each voltage — and **current-controlled** if it can be written $v = g(i)$. It may be both, either, or neither.



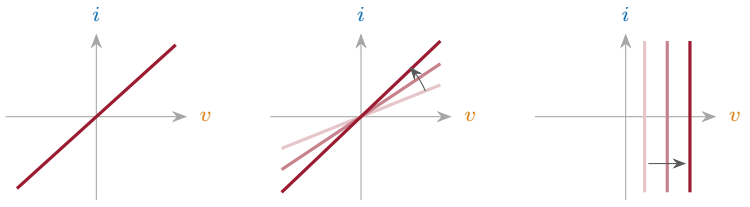
both | **v-controlled only** | **i-controlled only** | **neither**

For the tunnel diode the dashed horizontal line meets the curve three times: one voltage per current is impossible, so it is *not* current-controlled, while $i = f(v)$ is perfectly well defined. The glow tube is the mirror image. The ideal diode fails both tests: at $v = 0$ any positive current is allowed, and at $i = 0$ any negative voltage is.

7.5. Time-invariant

Time-invariant

The characteristic is **the same curve at every instant**. If it moves, stretches or tilts as t runs, the element is time-varying.



time-invariant | **time-varying: $R(t)$** | **time-varying: $v_s(t)$**

A potentiometer being turned and an AC wall socket are both time-varying resistors — the curve is still a curve, but a different one at each instant.

8. The gallery, characterized

Element	linear	bilateral	passive	v-contr.	i-contr.	time-inv.
linear resistor, $R > 0$	yes	yes	yes	yes	yes	yes
linear resistor, $R < 0$	yes	yes	no	yes	yes	yes
short circuit	yes	yes	yes	no	yes	yes
open circuit	yes	yes	yes	yes	no	yes
ideal diode	no	no	yes	no	no	yes
pn junction diode	no	no	yes	yes	yes	yes
tunnel diode	no	no	yes	yes	no	yes
glow tube	no	no	yes	no	yes	yes
DC voltage source, $v_s \neq 0$	no	no	no	no	yes	yes
DC current source, $i_s \neq 0$	no	no	no	yes	no	yes
AC voltage source	no	no	no	no	yes	no

Three rows are worth checking by eye rather than trusting.

Example 3.2 — reading three rows off the pictures

The short circuit is the line $v = 0$. Ask for $i = f(v)$: at $v = 0$ every current is allowed, so no function exists — not voltage-controlled. Ask for $v = g(i)$: $g(i) = 0$ for every i — current-controlled. The open circuit is the mirror image.

The *pn* diode, $i = I_s(e^{v/V_T} - 1)$, is a single-valued increasing function both ways round, so it is voltage- *and* current-controlled. Its curve stays in quadrants 1 and 3, so it is passive; it is bent, so nonlinear; and it is not symmetric about the origin, so not bilateral.

The DC voltage source is a vertical line at $v = v_s$. It is not linear (misses the origin), not bilateral, not passive (it reaches quadrant 4), not voltage-controlled (at $v = v_s$ any current is allowed) — but it is current-controlled, since $v = v_s$ for every i.

9. The LTI resistor

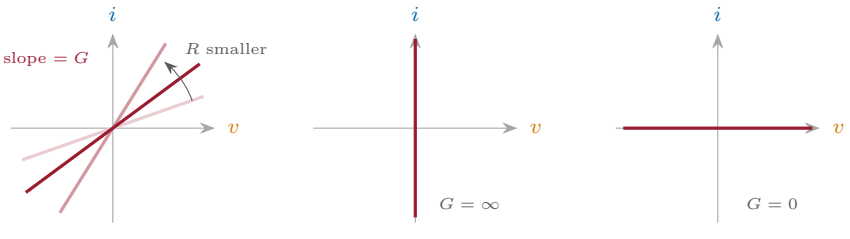
Of the five words above, take two: **linear** and **time-invariant**. Linear makes the characteristic a straight line through the origin; time-invariant stops that line from moving. Nothing is left to choose but the slope — and that is the element you have been calling a resistor since school.

LTI resistor

A resistor whose characteristic is **one fixed straight line through the origin**:

$$v = R\,i \qquad \Longleftrightarrow \qquad i = G\,v, \qquad G = \frac{1}{R}.$$

R is the **resistance**, in ohms ($\Omega = \mathrm{V/A}$); G is the **conductance**, in siemens ($\mathrm{S} = \mathrm{A/V}$).



$0 < R < \infty$ $R = 0$: **short circuit** $R = \infty$: **open circuit**

Turning the line from horizontal to vertical sweeps R from ∞ down to 0. The two end positions have names of their own — and we have now met them three times over: as sources of zero value, as two entries in the gallery, and here as the two limits of a resistance.

	R	G	**characteristic**	**what is fixed**
short circuit	0	∞	the i-axis, $v = 0$	the voltage
open circuit	∞	0	the v-axis, $i = 0$	the current

$G = 1/R$ is not always available

At the two extremes only one of the pair exists: a short circuit has $G = \infty$, an open circuit has $R = \infty$, and neither number is usable in a formula. This is the same fact as Section 7: a short circuit is current-controlled but not voltage-controlled, and an open circuit the other way round.

Example 3.3 — reading the line

A 100 Ω resistor has 5 V across it. Then

$$i = \frac{5}{100} = 50\,\text{mA}, \qquad G = \frac{1}{100} = 10\,\text{mS}.$$

Double the voltage and the current doubles: that is all "linear" means. Wait an hour and nothing about the line changes: that is "time-invariant".

10. Power in an LTI resistor

Substituting the terminal equation into $p = vi$ gives three ways to write the same thing:

$$\boxed{p \;=\; v\,i \;=\; R\,i^{2} \;=\; G\,v^{2} \;=\; \frac{v^{2}}{R}}$$

The middle forms are squares. So for $R > 0$,

$$p(t) \;\geq\; 0 \quad \text{at every instant,} \qquad w(t) = \int_{t_0}^{t} p(\tau)\,d\tau \;\geq\; 0 \quad \text{and never decreases.}$$

A resistor has no memory and no store

Energy enters and leaves as heat *at the same instant*. The graph of w only climbs; nothing the circuit does afterwards can bring that energy back. Keep this in mind — next lecture's capacitor behaves in exactly the opposite way.

Example 3.4 — a resistor as a heater

A 12 V source across 4 Ω:

$$i = 3\,\text{A}, \qquad p = 12 \times 3 = Ri^{2} = 4(3)^{2} = 36\,\text{W},$$

and in ten minutes $w = 36 \times 600 = 21.6\,\text{kJ}$, all of it heat. That is why resistors carry a power rating as well as a resistance: a $\frac{1}{4}$ W part in this position would not survive.

11. Why the picture was worth drawing

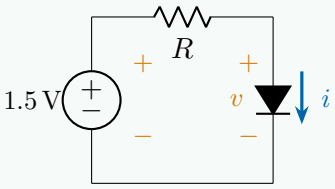
A diode has no resistance to write down — only a curve. That curve is still enough to solve a circuit, and doing it once shows why we bothered to draw characteristics at all.

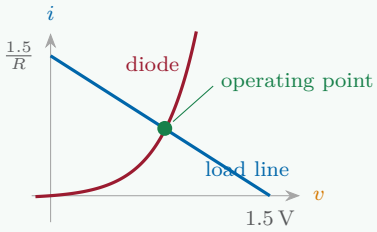
Example 3.5 — a diode driven by a source and a resistor

A 1.5 V source in series with R drives a diode. Walk round the loop with the diode voltage v and current i:

$$v = 1.5 - Ri \qquad \Longleftrightarrow \qquad i = \frac{1.5 - v}{R}.$$

That is a straight line in the same plane — the **load line**. The diode's own curve is the exponential. The circuit's answer is the single point on both.





Change R and the load line tilts about the point (1.5 V, 0); change the source and it slides sideways. The diode's curve never moves. No algebra was needed — and for a nonlinear element there often is no algebra to be had.

Summary

- A two-terminal element is completely described by its **characteristic**: the set of (v, i) pairs it permits, drawn as a curve in the v–i plane.
- **Independent voltage source**: vertical line $v = v_s(t)$, any current. **Independent current source**: horizontal line $i = i_s(t)$, any voltage. Neither one decides the other variable — the rest of the circuit does.
- Short circuit = voltage source with $v_s = 0$; open circuit = current source with $i_s = 0$.
- A **resistor** is any element whose characteristic is a curve — an algebraic relation, no memory. Diodes and sources are resistors; capacitors and inductors are not.
- Five questions about the shape of that curve: **linear** (straight, through the origin), **bilateral** (symmetric about the origin), **passive** (quadrants 1 and 3 only), **voltage-** or **current-controlled** (single-valued which way round?), **time-invariant** (does the curve move?).
- An **LTI resistor** is linear *and* time-invariant: one fixed straight line through the origin, $v = Ri$, $i = Gv$, $G = 1/R$. Ohms and siemens.
- $R = 0$ is a short circuit, $R = \infty$ an open circuit; at each end only one of R, G is a usable number. Its power $Ri^2 = Gv^2 \geq 0$ leaves as heat at the same instant — no memory, no store.
- Draw the element's curve and the rest of the circuit's **load line** in the same plane: where they cross is where the circuit operates. No algebra needed.

Next lecture

The capacitor — the first element that *remembers*. Its characteristic lives in a different plane altogether, and the energy you put into it comes back out.